

SUMMARY TABLE: PHYSICS GRADE 10

Sub-Strand 3.1: Radioactivity and Stability of Isotopes — Teacher Reference (Pre-filled)

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Sub-Strand	Sub-Strand 3.1: Radioactivity and Stability of Isotopes
Driving Question	DRIVING QUESTION: "Why do some atoms in Kenyan rocks spontaneously release invisible energy — and how can we harness or protect ourselves from that energy?"

INSTRUCTIONS

FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 The Invisible Killer in the Rock: Nuclear Structure, Isotopes & Launching the Phenomenon	A photographic film was exposed inside a sealed, lightproof box placed next to a sample of pitchblende (uranium ore) — with no light, heat, or sound present. The film showed a clear dark silhouette of the rock.	Atoms contain a nucleus made of protons and neutrons (nucleons). The atomic number (Z), mass number (A), and neutron number ($N = A - Z$) define a nuclide, written as AZX. Isotopes are atoms of the same element with the same Z but different N — chemically identical but potentially very different in nuclear stability. Stable nuclides cluster in the 'belt of stability' on an N vs Z graph; nuclides outside this belt (like Uranium-238 and Carbon-14) are unstable. Radioactivity is the spontaneous emission of energy or particles from an unstable nucleus — it requires no external energy input and cannot be stopped by ordinary chemical means.	The pitchblende rock exposes the film because it contains uranium nuclei (e.g., Uranium-238) whose N:Z ratio places them far outside the belt of stability. These unstable nuclei spontaneously release invisible energy and/or particles — radioactive emissions — as they attempt to reorganise into a more stable configuration. This is a nuclear process, not a chemical one, which is why the rock continues emitting radiation indefinitely without any external energy source. The type of emission and the rate of decay (half-life) are concepts we will investigate in the lessons that follow.
Lesson 2 Types of Radiation: Alpha, Beta, and Gamma — Penetrating Power and Nuclear Identity	Learners observed diagrams and data cards showing that alpha radiation is stopped by a sheet of paper, beta radiation is stopped by aluminium foil, and gamma radiation requires lead shielding. They observed that the radiation type with the greatest ionising ability (alpha) has the lowest penetrating power, and vice versa for gamma radiation. They also observed	Learners learned that radioactive nuclei emit three types of radiation — alpha (α : charge +2, mass ~4 amu, low penetration, high ionisation), beta (β : charge -1, mass ~0 amu, moderate penetration, moderate ionisation), and gamma (γ : charge 0, mass 0, high penetration, low ionisation). Alpha decay reduces mass number by 4 and atomic number by 2, moving the nucleus	Learners can now explain that the pitchblende rock continuously exposes photographic film because uranium-238 nuclei — which sit outside the belt of nuclear stability due to an excess of neutrons relative to protons — spontaneously emit alpha particles to reduce their mass and proton number, moving toward a more stable nuclear

	the uranium-238 → thorium-234 → protactinium-234 decay chain represented as arrows on an N vs Z stability plot.	diagonally toward the belt of stability on the N vs Z plot. Beta-minus decay increases atomic number by 1 and decreases neutron number by 1, shifting the nucleus diagonally (one step down and to the right) toward the belt. Gamma decay releases energy without changing nuclear composition. These decay types explain what invisible emissions are leaving the pitchblende rock and why.	configuration. The resulting daughter nuclei may still be unstable and emit beta particles to further adjust their neutron-to-proton ratio. Gamma rays carry away excess nuclear energy during these transitions. Each emission type differs in its ability to penetrate matter: alpha particles are stopped by a sheet of paper, making them less externally hazardous but extremely dangerous if inhaled (as in mining dust from Kenyan mineral-rich regions); beta particles require aluminium shielding; and gamma rays require lead or thick concrete, posing the greatest external radiation hazard to Kenyan communities near uranium-bearing geological formations.
Lesson 3 Detection of Radiation: Seeing the Invisible	Learners observed (via virtual simulation and video): (1) a Geiger-Müller counter producing clicks even with no radioactive source present (background radiation); (2) count rates increasing when alpha, beta, and gamma sources were brought near the GM tube, with different responses to shielding materials (paper, aluminium, lead); (3) cloud chamber tracks showing thick straight lines for alpha particles, thin curved lines for beta particles, and faint scattered trails for gamma radiation; (4) images and descriptions of GM-type detectors and dosimeters used at Kenyatta National Hospital (KNH) and environmental monitoring stations operated by the Kenya Nuclear Regulatory Authority (KNRA) near Mrima Hill, Kwale County.	Learners learned that: (1) a Geiger-Müller tube detects radiation because ionising radiation enters through a thin mica window, ionises argon gas atoms inside the tube, and the freed electrons accelerate toward a central anode under high voltage, producing an electrical pulse registered as a 'click'; (2) background radiation is always present in the environment — sourced from cosmic rays, naturally radioactive materials in building stone (e.g. Rift Valley granite), radon gas from soil, and trace radioisotopes in food (e.g. potassium-40 in sukuma wiki and githeri); (3) a cloud chamber makes radiation tracks visible by allowing particles to ionise supersaturated alcohol vapour, forming condensation trails whose shape reveals the particle type; (4) detection instruments are essential tools for radiation safety in Kenyan hospitals, mining regions, and agricultural research, allowing invisible nuclear emissions to be converted into actionable, quantitative data.	Learners explained that: (1) the invisible radiation from the pitchblende ore in the anchoring phenomenon — the same radiation that exposed photographic film in the dark — is detectable by a GM tube because it ionises matter (gas atoms in the tube) as it passes through, converting a nuclear event into an electrical signal; (2) different radiation types (α , β , γ) produce different GM count-rate responses with shielding because they differ in mass, charge, and ionising power — alpha particles are stopped by paper and air, beta particles by a few millimetres of aluminium, and gamma rays require dense lead shielding; (3) background radiation means that a GM tube will always record some counts even in the absence of a known source, so measurements of a radioactive sample must always be compared against the background count rate to determine the true emission from the source; (4) the rock in the anchoring phenomenon poses a quantifiable risk — detection devices give the numerical count rates that allow scientists, doctors, and regulators at KNRA and KNH to set safe exposure limits and protect Kenyan communities.
Lesson 4	When we rolled 100 dice and removed every 6, the number of surviving dice	The half-life of a radioactive isotope is the time for half the undecayed nuclei in a	The pitchblende rock has been emitting radiation for billions of years without running

Half-Life and Decay Curves: How Long Before the Rock Goes Quiet?	decreased each round, producing a curved graph that flattened out over time rather than a straight line. Every group extracted a similar half-life of approximately 3 to 4 rolls despite different individual data patterns. We also observed that the number of dice removed per round decreased as the surviving population got smaller.	sample to decay. Decay follows an exponential pattern described by $N = N_0 \times (\text{one-half})^n$. Different isotopes have vastly different half-lives: Carbon-14 has a half-life of 5730 years (used to date the Olorgesailie charcoal) and Iodine-131 has a half-life of 8 days (used in KNH thyroid treatment). Uranium-238 has a half-life of 4.5 billion years, which explains why the pitchblende rock in our phenomenon is still emitting radiation today.	out because uranium-238 has an enormously long half-life. Only a tiny fraction of its nuclei have decayed since Earth formed, leaving vast numbers of undecayed nuclei still present. The decay process is random at the level of individual nuclei but statistically predictable for large populations — exactly as our dice simulation showed. The rate of emission is not constant forever (it slowly decreases as nuclei are used up) but on any human timescale the rock appears to emit radiation indefinitely.
Lesson 5 Nuclear Equations: Writing and Balancing Radioactive Decay	Learners observed: (1) a fully worked α-decay equation ($^{238}_{92}\text{U} \rightarrow ^{234}_{90}\text{Th} + ^4_2\text{He}$) demonstrated step-by-step on the board, showing conservation of nucleon number and proton number; (2) three guided equation cards covering α-decay (Radium in pitchblende), β-decay (Carbon-14 dating at Fort Jesus, Mombasa), and γ-emission (Cobalt-60 at KNH Nairobi), each requiring the daughter nucleus to be identified using a periodic table; (3) an interactive simulation (ARES offline module) in which parent nuclei were input and daughter nuclei and emitted particles appeared, with ΔA and ΔZ values recorded in a table.	Learners learned that: (1) nuclear equations must conserve nucleon number (A) and proton number (Z) on both sides — these are the fundamental rules governing all radioactive decay; (2) α-decay always reduces A by 4 and Z by 2, producing a daughter nucleus two elements lower on the periodic table; (3) β-minus decay keeps A constant while increasing Z by 1, producing a daughter nucleus one element higher; (4) γ-emission changes neither A nor Z — it releases energy only; (5) each decay can be plotted as a directional arrow on the N vs Z graph, and every arrow points toward the belt of stability, confirming that radioactive decay is the nucleus's way of achieving a more stable configuration.	Learners explained: Using balanced nuclear equations and the N vs Z stability plot, learners explained that the pitchblende ore exposes photographic film in complete darkness because $^{238}_{92}\text{U}$ nuclei — sitting outside the belt of stability — spontaneously eject alpha particles (^4_2He), transforming into $^{234}_{90}\text{Th}$ daughter nuclei that sit closer to the belt of stability. The emitted alpha particles (and subsequent beta particles and gamma rays from further decays in the chain) carry enough energy to penetrate the dark container and interact with the photographic film's silver halide crystals, leaving a visible exposure. Conservation of nucleon number and proton number governs every step, making the invisible process mathematically predictable and verifiable.
Lesson 6 Nuclear Fission and Fusion: Unlocking the Energy Inside the Rock	Using the PhET Nuclear Fission simulation, learners observed: a U-235 nucleus splitting into two fission fragments + 2–3 neutrons + energy when struck by a neutron; a chain reaction branching diagram showing exponential growth from one initial neutron; the role of control rods in moderating the chain reaction; and the full energy transformation pathway inside a nuclear reactor (nuclear $\rightarrow$ thermal $\rightarrow$ kinetic $\rightarrow$ electrical energy). From the video, learners also observed a conceptual explanation of nuclear fusion and the KNEB context for Kenya's nuclear energy plans.	Learners learned that: (1) Nuclear fission is the splitting of a heavy nucleus (e.g., U-235) by a neutron, producing fission fragments, more neutrons, and large amounts of energy; (2) Nuclear fusion is the joining of light nuclei (e.g., hydrogen isotopes) under extreme conditions, releasing even greater energy per kilogram; (3) In both reactions, the products have slightly less mass than the reactants — this 'missing' mass (mass defect) is converted to energy via $E = mc^2$, explaining WHERE the energy in the pitchblende rock ultimately comes from; (4) A controlled chain reaction	Learners constructed evidence-based explanations that: (1) the energy released when our pitchblende rock emits an alpha or gamma ray comes from the conversion of a tiny amount of nuclear mass into energy (mass defect $\rightarrow E = mc^2$), meaning the source of the radiation is literally the mass of the uranium nucleus itself; (2) nuclear fission and fusion are engineered applications of the same nuclear energy principle — deliberately triggering nuclear rearrangement to release large amounts of energy rapidly; (3) Kenya's decision about nuclear power involves weighing the benefit

		in a nuclear reactor generates electricity, while an uncontrolled chain reaction produces a nuclear explosion — control rods manage this; (5) Kenya (through KNEB) is actively considering nuclear power as part of its energy mix alongside geothermal (Olkaria), hydro, and solar sources.	of large-scale, low-carbon electricity generation against risks of nuclear waste, safety incidents, and high upfront costs — a decision that requires scientific literacy, not just opinion.
Lesson 7 Applications, Safety, and Unit Review: Closing the Case of the Invisible Killer	Learners observed four real-world Kenyan applications of radioactivity through the applications carousel: cobalt-60 radiotherapy at Kenyatta National Hospital for cervical cancer treatment; KALRO gamma irradiation of stored maize, beans, and sorghum for pest control and crop improvement in counties like Makueni; beta-source industrial thickness gauging in manufacturing; and carbon-14 versus uranium-lead dating applied to the Homo habilis fossil at Koobi Fora, Lake Turkana. Learners also observed their own Lesson 1 naive models alongside their final cumulative models, making the arc of conceptual change visible.	Learners learned that the same nuclear instability that caused the pitchblende rock to expose photographic film in Lesson 1 is the foundational phenomenon behind life-saving medical treatments, improved crop yields for Kenyan farmers, precise industrial quality control, and the dating of Kenya's extraordinary hominin fossil record. They learned that radiation safety is governed by the ALARA principle — minimising exposure through shielding (lead, concrete), maximising distance (inverse square law: doubling distance reduces dose to one-quarter), and minimising time — and that the half-life of an isotope determines both its usefulness and the duration of its hazard. They consolidated the complete unit explanatory framework: unstable nucleus (imbalanced neutron-to-proton ratio) → spontaneous decay (α , β , or γ) → nuclear equation → half-life → penetrating radiation → detectable, harness-able, and manageable energy.	Learners explained, in their own words and in writing, why the pitchblende rock spontaneously and continuously exposes photographic film without any external energy input: uranium-238 nuclei have a neutron-to-proton ratio outside the belt of stability, causing spontaneous alpha decay (and a subsequent decay chain including beta and gamma emissions), releasing penetrating radiation that passes through the film container. The process continues indefinitely on human timescales because U-238 has a half-life of approximately 4.5 billion years, meaning only an infinitesimally small fraction of nuclei decay in any observable period — yet the number of atoms in even a small ore sample is so enormous (on the order of 10^{23}) that the count rate remains measurable and persistent. Learners also explained that this same physics, properly understood and controlled, enables Kenya to treat cancer patients, protect stored food, ensure product quality, and read the deep history of human origins from the rocks of the Rift Valley.

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